DHRUV MAKWANA

Languages: English, Gujarati, Hindi
Current Location: London, UK
Website: cst.cam.ac.uk/dcm41
References: Prof. Neel Krishnaswami, Prof. Peter Sewell

PHD, COMPUTER SCIENCE, 2020 – PRESENT

M.ENG IN COMPUTER SCIENCE (WITH PSYCHOLOGY), 2014 – 2018

TRINITY COLLEGE, CAMBRIDGE

I am open to any work at the intersection of elegant abstractions, technical infrastructure problems and readable, working code. So far, I have focused on tooling performance [C,G,M], types [E], and verification [A,B]. I'm enthusiastic about supporting others: I've trained new hires and mentored interns [H], given talks [C,G], made educational videos [P], and taught undergrads [O,R,S] and children [Q].

Research

A. PhD Thesis (full draft): *On the Theory and Engineering of Verifying Systems C*
B. POPL'23 – *CN: Verifying Systems C Code …* and POPL'25 – *Fulminate: Testing … Specifications in C*
C. ECOOP 2019 – *NumLin: Linear Types for Linear Algebra*
D. Report – *Implementing Balanced Polling for OCaml* (explored safe-points for multicore OCaml)

Employment History

University of Cambridge | PhD Student in Computer Science | Apr 2020 – Present

Meta, London | PhD Software Eng Intern, Hack Type-checker, Dev Infra | Jul – Sep 2023

E. Reduced risk of disruption to www.facebook.com with stricter switch check in Hack (GitHub)
F. Result: improved robustness of 98.5% of 170K switches; signaled potential errors in rest

Goldman Sachs, London | Analyst (Software Developer), SecDB Architecture | Jun 2018 – Mar 2020

G. Explored feasibility of Slang running on Truffle/GraalVM: talk at Curry On (2019)
H. Supervised intern project (gRPC for Slang) and training (Java 8, React/Redux and Slang/SecDb)
I. Interviewed candidates & improved hiring (updated job spec, encouraged work-representative questions)

Arm, Cambridge | Verification Engineer, CPU Group | Jun – Aug 2017

J. Set-up a new workflow for model-checking undefined decoders
K. Verified (model-checking) undefined decoders on two released processors for two different architectures

Open-Source Contributions

L. CN and Cerberus: memory model, UX improvements (errors, LSP, VS Code), refactored parser
M. c-tree-carver: Clang-based tree carving tool for C/C++
N. (Ongoing) Raven ML framework for OCaml: implementation of einsum (GitHub)

Teaching

O. Undergrad supervisions: OCaml, Discrete Maths, Java, C/C++, Prolog, Compilers, Types, Semantics, Complexity T.
P. YouTube video series (ATypical CompSci) to teach functional programming, targeted at 1st years
Q. CoderDojo – programming in Scratch and Python for young children and teenagers
R. Global Wellbeing reading group: created and ran two rounds, inspired spin-off courses
S. Aurelius Society: taught Stoic philosophy via discussion groups, socials, and talks

Activities

Within the Plant-based Cambridge campaign, I organized a talk with 30K online views and led Divestment work. At the Aurelius Foundation, I speak and write on Stoicism as a Youth Ambassador. As treasurer of Cambridge University Bollywood Dance Troupe, I recovered £450 in lost funds. I volunteer with local animal rights groups and enjoy cooking (specializing in vegan Gujarati cuisine).